Examiner only

5. Mean speed is defined by the equation:

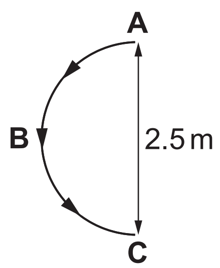
$$\text{mean speed} = \frac{\text{total distance}}{\text{total time}}$$

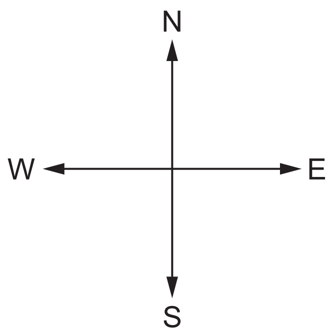
(a) State the definition of mean **velocity**. [1]

....

....

(b) Ieuan and Seren play with their toy train. A train takes 4.0 s to travel at constant speed in a semicircle **ABC** as shown.





(i) Show that the train's speed is approximately $1\,\text{m}\,\text{s}^{-1}$. [1]

....

....

(ii) Calculate the mean velocity of the train as it moves from **A** to **C**. [2]

....

....

....

(iii) Ieuan believes that, since the speed between **A** and **C** is constant, the train is not accelerating. Seren disagrees. Explain who is correct, justifying your answer. [3]

....

....

....

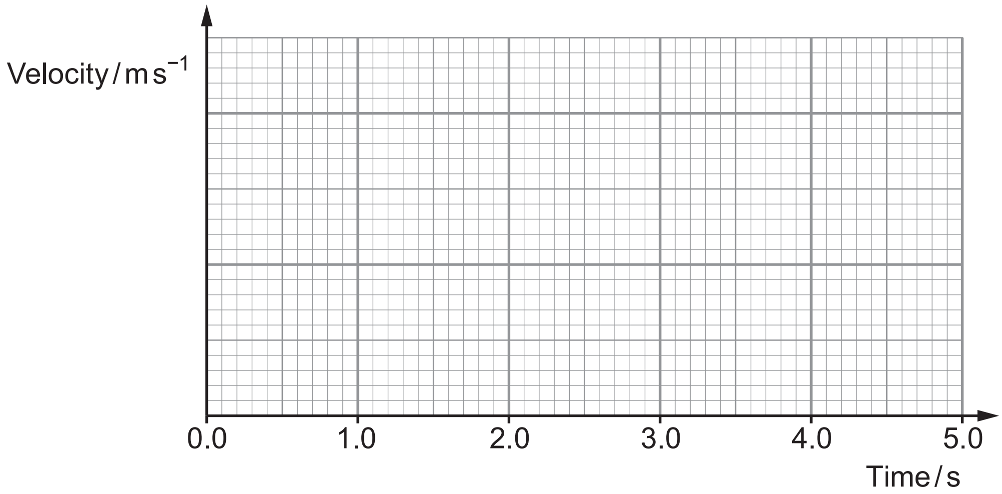
....

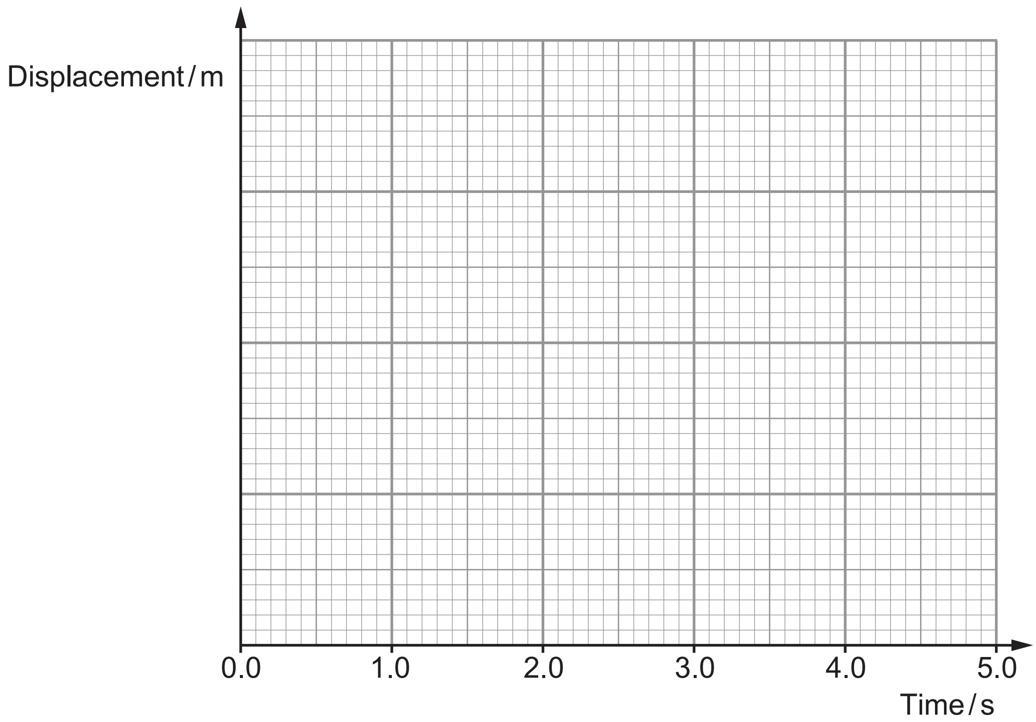
....

....

Examiner only

(c) **From C**, the train continues in a straight line with a speed of $1.0\,\mathrm{m\,s^{-1}}$ for 2.0 s. It then decelerates uniformly to rest in a distance of 1.2 m. Complete the velocity-time graph and the displacement-time graph **for the motion from C** on the grids below.
Space for calculations. [6]





13